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|------------------------------|---------------------|
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| Section: COM242ML            | Assignment #: 1     |

### Training Data (Given)

| Student | Hours Studied $X_1$ | Sleep Hours $X_2$ | Pass/Fail (Y) |
|---------|---------------------|-------------------|---------------|
| 1       | 1.0                 | 5.0               | 0 (Fail)      |
| 2       | 2.0                 | 5.5               | 0 (Fail)      |
| 3       | 3.0                 | 6.0               | 0 (Fail)      |
| 4       | 4.5                 | 5.0               | 0 (Fail)      |
| 5       | 5.0                 | 6.5               | 1 (Pass)      |
| 6       | 5.5                 | 7.0               | 1 (Pass)      |
| 7       | 6.0                 | 6.0               | 1 (Pass)      |
| 8       | 7.0                 | 7.0               | 1 (Pass)      |
| 9       | 8.0                 | 6.0               | 1 (Pass)      |
| 10      | 9.0                 | 7.5               | 1 (Pass)      |

### Computed Distances using Euclidean Distance (4, 6)

| Student | Hours Studied (X1) | Sleep Hours (X2) | Pass/Fail (Y) | Euclidean Distance |
|---------|--------------------|------------------|---------------|--------------------|
| 1       | 1.0                | 5.0              | 0             | 3.16               |
| 2       | 2.0                | 5.5              | 0             | 2.06               |
| 3       | 3.0                | 6.0              | 0             | 1                  |
| 4       | 4.5                | 5.0              | 0             | 1.11               |
| 5       | 5.0                | 6.5              | 1             | 1.11               |
| 6       | 5.5                | 7.0              | 1             | 1.80               |
| 7       | 6.0                | 6.0              | 1             | 2                  |
| 8       | 7.0                | 7.0              | 1             | 3.16               |
| 9       | 8.0                | 6.0              | 1             | 4                  |
| 10      | 9.0                | 7.5              | 1             | 5.22               |
| New     | 4.0                | 6.0              |               |                    |

### 3 Nearest Neighbors

| Student | Hours Studied (X1) | Sleep Hours (X2) | Pass/Fail (Y) | Euclidean Distance |
|---------|--------------------|------------------|---------------|--------------------|
| 1       | 1.0                | 5.0              | 0             | 3.16               |
| 2       | 2.0                | 5.5              | 0             | 2.06               |
| 3       | 3.0                | 6.0              | 0             | 1                  |
| 4       | 4.5                | 5.0              | 0             | 1.11               |
| 5       | 5.0                | 6.5              | 1             | 1.11               |
| 6       | 5.5                | 7.0              | 1             | 1.80               |
| 7       | 6.0                | 6.0              | 1             | 2                  |
| 8       | 7.0                | 7.0              | 1             | 3.16               |
| 9       | 8.0                | 6.0              | 1             | 4                  |
| 10      | 9.0                | 7.5              | 1             | 5.22               |
| New     | 4.0                | 6.0              |               |                    |

### Majority Vote

| Student | Hours Studied (X1) | Sleep Hours (X2) | Pass/Fail (Y) | Euclidean Distance |      |
|---------|--------------------|------------------|---------------|--------------------|------|
| 1       | 1.0                | 5.0              | 0             | 3.16               |      |
| 2       | 2.0                | 5.5              | 0             | 2.06               |      |
| 3       | 3.0                | 6.0              | 0             | 1                  | Fail |
| 4       | 4.5                | 5.0              | 0             | 1.11               | Fail |
| 5       | 5.0                | 6.5              | 1             | 1.11               | Pass |
| 6       | 5.5                | 7.0              | 1             | 1.80               |      |
| 7       | 6.0                | 6.0              | 1             | 2                  |      |
| 8       | 7.0                | 7.0              | 1             | 3.16               |      |
| 9       | 8.0                | 6.0              | 1             | 4                  |      |
| 10      | 9.0                | 7.5              | 1             | 5.22               |      |
| New     | 4.0                | 6.0              |               |                    |      |

*Fail = 2, Pass = 1*

**Prediction:** A new student who studied 4 hours and slept 6 hours will **fail** the exam.

### Discussions Answers:

1. The final prediction was that the new student who studied 4 hours and slept 6 hours will fail the exam.
2. The prediction would be less error and accurate if we used  $k = 5$ .